Name Aayush

Roll NO.:2401010042

Section-B

Course:B.tech Cse Core

Assignment 3 :

❖ Aim

To design and develop a full-stack Quiz Master web application using HTML, CSS, Python, and Django framework, enabling users to attempt multiple-choice questions and view their scores with feedback.

❖ Tasks

- Set up a Django project and application
- Create quiz questions and store them (static or database)
- Design HTML templates for Home, Quiz, and Result pages
- Implement backend logic using Django views
- Handle form submission and user responses
- Calculate quiz score based on answers
- Display result and feedback to the user
- Apply CSS styling for better user interface

❖ Technologies Used

- **Frontend:** HTML, CSS
- **Backend:** Python, Django
- **Framework:** Django Web Framework
- **Tools:** VS Code / PyCharm, Web Browser
- **Version Control (Optional):** Git

❖ Code :

➤ **HTML CODE :**

❖ **Index.html**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Quiz Master</title>
  {% load static %}
  <link rel="stylesheet" href="{% static 'quiz/style.css' %}">
</head>
<body>
  <div class="page home-page">
    <div class="home-card">
      <div class="logo">🧠</div>
      <h1>Quiz Master</h1>
      <p class="subtitle">Web Programming with Python & Django</p>
      <div class="info-grid">
        <div class="info-box">
          <span class="info-num">7</span>
          <span class="info-label">Questions</span>
        </div>
        <div class="info-box">
          <span class="info-num">MCQ</span>
          <span class="info-label">Format</span>
        </div>
        <div class="info-box">
          <span class="info-num">10</span>
```

```
                <span class="info-label">Minutes</span>
            </div>
        </div>
        <a href="{% url 'quiz' %}" class="btn-start">Start Quiz →</a>
    </div>
  </div>
</body>
</html>
```

❖ Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Quiz - Quiz Master</title>
  {% load static %}
  <link rel="stylesheet" href="{% static 'quiz/style.css' %}">
</head>
<body>
  <div class="page quiz-page">
    <div class="quiz-container">
      <div class="quiz-header">
        <h1>🧠 Quiz Master</h1>
        <div class="timer-box">
          ⏱️ Time: <span id="timer">10:00</span>
        </div>
      </div>
```

```
        <form method="POST" action="{% url 'result' %}" id="quiz-form">
          {% csrf_token %}

          {% for q in questions %}
          <div class="question-card" id="card-{{ forloop.counter }}">
            <div class="q-number">Question {{ forloop.counter }} of
{{ questions|length }}</div>
            <p class="q-text">{{ q.question }}</p>
            <div class="options">
              {% for option in q.options %}
              <label class="option-label">
                <input type="radio" name="q{{ q.id }}" value="{{ forloop.counter0 }}"
required>
                <span class="option-text">{{ option }}</span>
              </label>
              {% endfor %}
            </div>
          </div>
          {% endfor %}

          <div class="submit-area">
            <p class="submit-note">⚠ Make sure you answer all questions before
submitting.</p>
            <button type="submit" class="btn-submit">Submit Quiz ✓</button>
          </div>
        </form>
      </div>
    </div>

    <script>
```

```
    // Timer
    let seconds = 600;
    const timerEl = document.getElementById('timer');
    const interval = setInterval(() => {
      seconds--;
      const m = Math.floor(seconds / 60).toString().padStart(2, '0');
      const s = (seconds % 60).toString().padStart(2, '0');
      timerEl.textContent = `${m}:${s}`;
      if (seconds <= 60) timerEl.style.color = '#e74c3c';
      if (seconds <= 0) {
        clearInterval(interval);
        document.getElementById('quiz-form').submit();
      }
    }, 1000);

    // Highlight selected option
    document.querySelectorAll('.option-label').forEach(label => {
      label.querySelector('input').addEventListener('change', () => {
        label.closest('.options').querySelectorAll('.option-label').forEach(l =>
l.classList.remove('selected'));
        label.classList.add('selected');
      });
    });
  </script>
</body>
</html>
```

❖ result.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Results - Quiz Master</title>
  {% load static %}
  <link rel="stylesheet" href="{% static 'quiz/style.css' %}">
</head>
<body>
  <div class="page result-page">
    <div class="result-container">
      <div class="score-card">
        <div class="grade grade-{{ grade_color }}">{{ grade }}</div>
        <h1>Your Score</h1>
        <div class="score-display">{{ score }} / {{ total }}</div>
        <div class="progress-bar">
          <div class="progress-fill" style="width: {{ percent }}%"></div>
        </div>
        <p class="percent">{{ percent }}%</p>
        <p class="feedback">{{ feedback }}</p>
        <div class="result-btns">
          <a href="{% url 'quiz' %}" class="btn-restart">🔁 Restart Quiz</a>
          <a href="{% url 'home' %}" class="btn-home">🏠 Home</a>
        </div>
      </div>

      <div class="review-section">
```

```
<h2>Review Answers</h2>
{% for r in results %}
<div class="review-card {% if r.is_correct %}correct{% else %}wrong{% endif %}">
  <p class="review-q"><strong>Q{{ forloop.counter }}.</strong> {{ r.question }}</p>
  <div class="review-options">
    {% for opt in r.options %}
    <div class="review-option
      {% if forloop.counter0 == r.correct %}correct-opt{% endif %}
      {% if forloop.counter0 == r.user_answer and not r.is_correct %}wrong-opt{% endif %}
    ">
      {% if forloop.counter0 == r.correct %}✓{% elif forloop.counter0 == r.user_answer %}✗{% else %}○{% endif %}
      {{ opt }}
    </div>
    {% endfor %}
  </div>
  <p class="review-status">
    {% if r.is_correct %}
      ☑ Correct!
    {% elif r.user_answer is None %}
      ⚠ Not answered
    {% else %}
      ✕ Wrong — Correct: {{ r.correct_text }}
    {% endif %}
  </p>
</div>
{% endfor %}
</div>
```

```
      </div>
    </div>
</body>
</html>
```

❖ style.css

```
* { margin: 0; padding: 0; box-sizing: border-box; }
:root {
    --primary: #4f46e5;
    --primary-dark: #3730a3;
    --accent: #06b6d4;
    --green: #10b981;
    --red: #ef4444;
    --orange: #f59e0b;
    --bg: #f1f5f9;
    --card: #ffffff;
    --text: #1e293b;
    --muted: #64748b;
    --border: #e2e8f0;
    --radius: 12px;
    --shadow: 0 4px 20px rgba(0,0,0,0.08);
}
body {
    font-family: 'Segoe UI', sans-serif;
    background: var(--bg);
    color: var(--text);
    min-height: 100vh;
```

```
}
.page {
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: flex-start;
    padding: 30px 20px;
}
.home-page {
    align-items: center;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
}
.home-card {
    background: white;
    border-radius: 20px;
    padding: 50px 40px;
    text-align: center;
    max-width: 420px;
    width: 100%;
    box-shadow: 0 20px 60px rgba(0,0,0,0.2);
}
.logo {
    font-size: 64px;
    margin-bottom: 10px;
    animation: bounce 2s infinite;
}
@keyframes bounce {
    0%, 100% { transform: translateY(0); }
    50% { transform: translateY(-10px); }
```

```
}
.home-card h1 {
    font-size: 2.2rem;
    color: var(--primary);
    margin-bottom: 8px;
}
.subtitle {
    color: var(--muted);
    margin-bottom: 30px;
    font-size: 0.95rem;
}
.info-grid {
    display: flex;
    gap: 16px;
    justify-content: center;
    margin-bottom: 32px;
}
.info-box {
    background: var(--bg);
    border-radius: var(--radius);
    padding: 16px 20px;
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 4px;
    flex: 1;
}
.info-num {
    font-size: 1.4rem;
```

```
    font-weight: 700;
    color: var(--primary);
}
.info-label {
    font-size: 0.75rem;
    color: var(--muted);
    text-transform: uppercase;
    letter-spacing: 0.5px;
}
.btn-start {
    display: inline-block;
    background: var(--primary);
    color: white;
    text-decoration: none;
    padding: 14px 40px;
    border-radius: 50px;
    font-size: 1.1rem;
    font-weight: 600;
    transition: all 0.2s;
}
.btn-start:hover {
    background: var(--primary-dark);
    transform: translateY(-2px);
    box-shadow: 0 8px 20px rgba(79, 70, 229, 0.4);
}
.quiz-page {
    background: var(--bg);
    padding-top: 20px;
}
```

```
.quiz-container {
    width: 100%;
    max-width: 720px;
}
.quiz-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background: var(--primary);
    color: white;
    padding: 16px 24px;
    border-radius: var(--radius);
    margin-bottom: 24px;
    box-shadow: var(--shadow);
}
.quiz-header h1 {
    font-size: 1.4rem;
}
.timer-box {
    background: rgba(255,255,255,0.2);
    padding: 8px 16px;
    border-radius: 20px;
    font-weight: 700;
    font-size: 1rem;
}
.question-card {
    background: var(--card);
    border-radius: var(--radius);
```

```
    padding: 24px;
    margin-bottom: 20px;
    box-shadow: var(--shadow);
    border-left: 4px solid var(--primary);
}
.q-number {
    font-size: 0.8rem;
    color: var(--muted);
    text-transform: uppercase;
    letter-spacing: 1px;
    margin-bottom: 10px;
    font-weight: 600;
}
.q-text {
    font-size: 1.1rem;
    font-weight: 600;
    margin-bottom: 18px;
    line-height: 1.5;
}
.options {
    display: flex;
    flex-direction: column;
    gap: 10px;
}
.option-label {
    display: flex;
    align-items: center;
    gap: 12px;
    padding: 12px 16px;
```

```
    border: 2px solid var(--border);
    border-radius: 8px;
    cursor: pointer;
    transition: all 0.15s;
}
.option-label:hover {
    border-color: var(--primary);
    background: #f5f3ff;
}

.option-label.selected {
    border-color: var(--primary);
    background: #ede9fe;
}

.option-label input[type="radio"] {
    accent-color: var(--primary);
    width: 18px;
    height: 18px;
    flex-shrink: 0;
}

.option-text {
    font-size: 0.97rem;
}

.submit-area {
    text-align: center;
    padding: 20px 0 40px;
```

```
}

.submit-note {
    color: var(--muted);
    font-size: 0.9rem;
    margin-bottom: 16px;
}

.btn-submit {
    background: var(--green);
    color: white;
    border: none;
    padding: 14px 48px;
    font-size: 1.1rem;
    font-weight: 700;
    border-radius: 50px;
    cursor: pointer;
    transition: all 0.2s;
}
.btn-submit:hover {
    background: #059669;
    transform: translateY(-2px);
    box-shadow: 0 8px 20px rgba(16, 185, 129, 0.4);
}
.result-page {
    background: var(--bg);
}

.result-container {
```

```
    width: 100%;
    max-width: 720px;
}

.score-card {
    background: var(--card);
    border-radius: 20px;
    padding: 40px;
    text-align: center;
    box-shadow: var(--shadow);
    margin-bottom: 30px;
}

.grade {
    display: inline-block;
    width: 80px;
    height: 80px;
    border-radius: 50%;
    font-size: 2rem;
    font-weight: 800;
    line-height: 80px;
    margin-bottom: 16px;
    color: white;
}

.grade-green { background: var(--green); }
.grade-blue { background: var(--primary); }
.grade-orange { background: var(--orange); }
.grade-red { background: var(--red); }
```

```
.score-card h1 {
    font-size: 1.3rem;
    color: var(--muted);
    margin-bottom: 8px;
}

.score-display {
    font-size: 3rem;
    font-weight: 800;
    color: var(--text);
    margin-bottom: 16px;
}

.progress-bar {
    background: var(--border);
    border-radius: 50px;
    height: 12px;
    margin: 0 auto 10px;
    max-width: 300px;
    overflow: hidden;
}

.progress-fill {
    height: 100%;
    background: linear-gradient(90deg, var(--primary), var(--accent));
    border-radius: 50px;
    transition: width 1s ease;
}
```

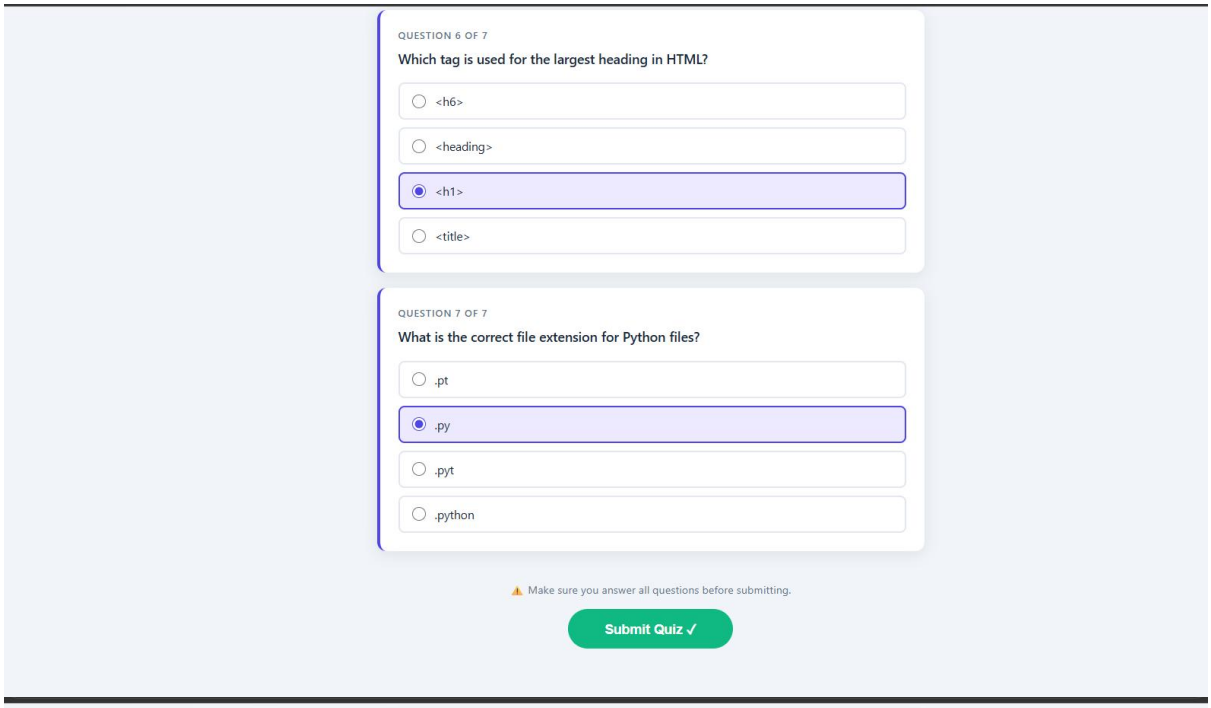
```
.percent {
    font-size: 1.5rem;
    font-weight: 700;
    color: var(--primary);
    margin-bottom: 10px;
}

.feedback {
    font-size: 1.1rem;
    color: var(--muted);
    margin-bottom: 24px;
}

.result-btns {
    display: flex;
    gap: 12px;
    justify-content: center;
    flex-wrap: wrap;
}

.btn-restart, .btn-home {
    text-decoration: none;
    padding: 12px 28px;
    border-radius: 50px;
    font-weight: 600;
    font-size: 0.95rem;
    transition: all 0.2s;
}
```

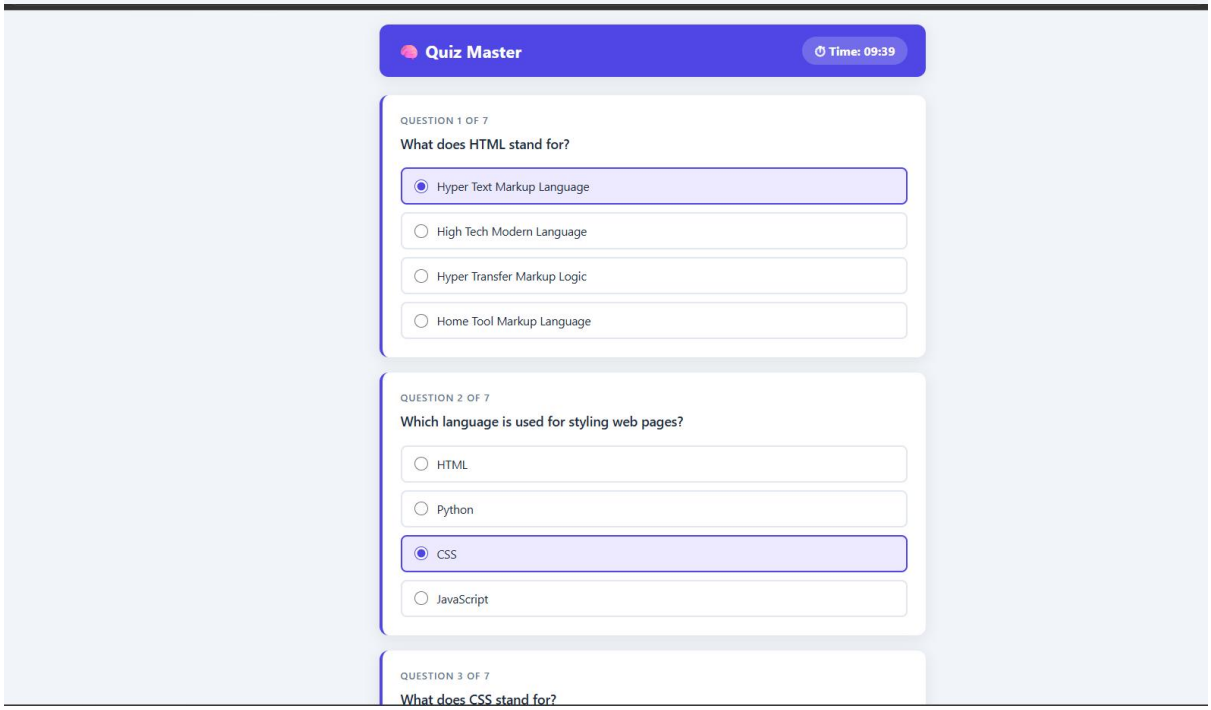
```
.btn-restart {
    background: var(--primary);
    color: white;
}

.btn-restart:hover {
    background: var(--primary-dark);
    transform: translateY(-2px);
}

.btn-home {
    background: var(--bg);
    color: var(--text);
    border: 2px solid var(--border);
}

.btn-home:hover {
    border-color: var(--primary);
    color: var(--primary);
}

/* ===== REVIEW SECTION ===== */
.review-section h2 {
    font-size: 1.3rem;
    margin-bottom: 16px;
    color: var(--text);
}
```

```
.review-card {
  background: var(--card);
  border-radius: var(--radius);
  padding: 20px;
  margin-bottom: 14px;
  box-shadow: var(--shadow);
  border-left: 4px solid var(--border);
}

.review-card.correct { border-left-color: var(--green); }
.review-card.wrong { border-left-color: var(--red); }

.review-q {
  margin-bottom: 12px;
  font-size: 0.97rem;
  line-height: 1.5;
}

.review-options {
  display: flex;
  flex-direction: column;
  gap: 6px;
  margin-bottom: 10px;
}

.review-option {
  padding: 8px 12px;
  border-radius: 6px;
  font-size: 0.9rem;
```

```
    background: var(--bg);
    color: var(--muted);
}

.review-option.correct-opt {
    background: #d1fae5;
    color: #065f46;
    font-weight: 600;
}

.review-option.wrong-opt {
    background: #fee2e2;
    color: #991b1b;
    font-weight: 600;
}

.review-status {
    font-size: 0.88rem;
    font-weight: 600;
    margin-top: 6px;
}
@media (max-width: 480px) {
    .home-card { padding: 36px 24px; }
    .quiz-header { flex-direction: column; gap: 10px; text-align: center; }
    .info-grid { flex-direction: column; }
    .score-display { font-size: 2.5rem; }
}
```

❖ Output


QUESTION 6 OF 7
Which tag is used for the largest heading in HTML?
<h6>
<heading>
<h1>
<title>
QUESTION 7 OF 7
What is the correct file extension for Python files?
.pt
.py
.pyt
.python
Make sure you answer all questions before submitting.
Submit Quiz ✓


Quiz Master
Time: 09:39
QUESTION 1 OF 7
What does HTML stand for?
Hyper Text Markup Language
High Tech Modern Language
Hyper Transfer Markup Logic
Home Tool Markup Language
QUESTION 2 OF 7
Which language is used for styling web pages?
HTML
Python
CSS
JavaScript
QUESTION 3 OF 7
What does CSS stand for?

A

Your Score

7 / 7

100%

Excellent! Outstanding performance!

Restart Quiz

Home

Review Answers

Q1. What does HTML stand for?

✓ Hyper Text Markup Language

○ High Tech Modern Language

○ Hyper Transfer Markup Logic

○ Home Tool Markup Language

Correct!

Q2. Which language is used for styling web pages?